



Artificial Intelligence

Project 1

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of an Artificial Intelligence Engineer at DecodeLabs. Project 1 is your foundation phase: The Rule-Based AI Chatbot. This track isn't about "deep learning" just yet—it's about Control Flow and Logic. Before you build systems that learn on their own, you must master the art of teaching a machine through explicit if-else instructions. By completing this milestone, you are proving you can create a continuous digital loop that simulates basic human interaction through pure programmatic decision-making. Let's design the fundamental architecture of your first intelligent interface with absolute clarity.



GUIDELINES:

Qualification Criteria

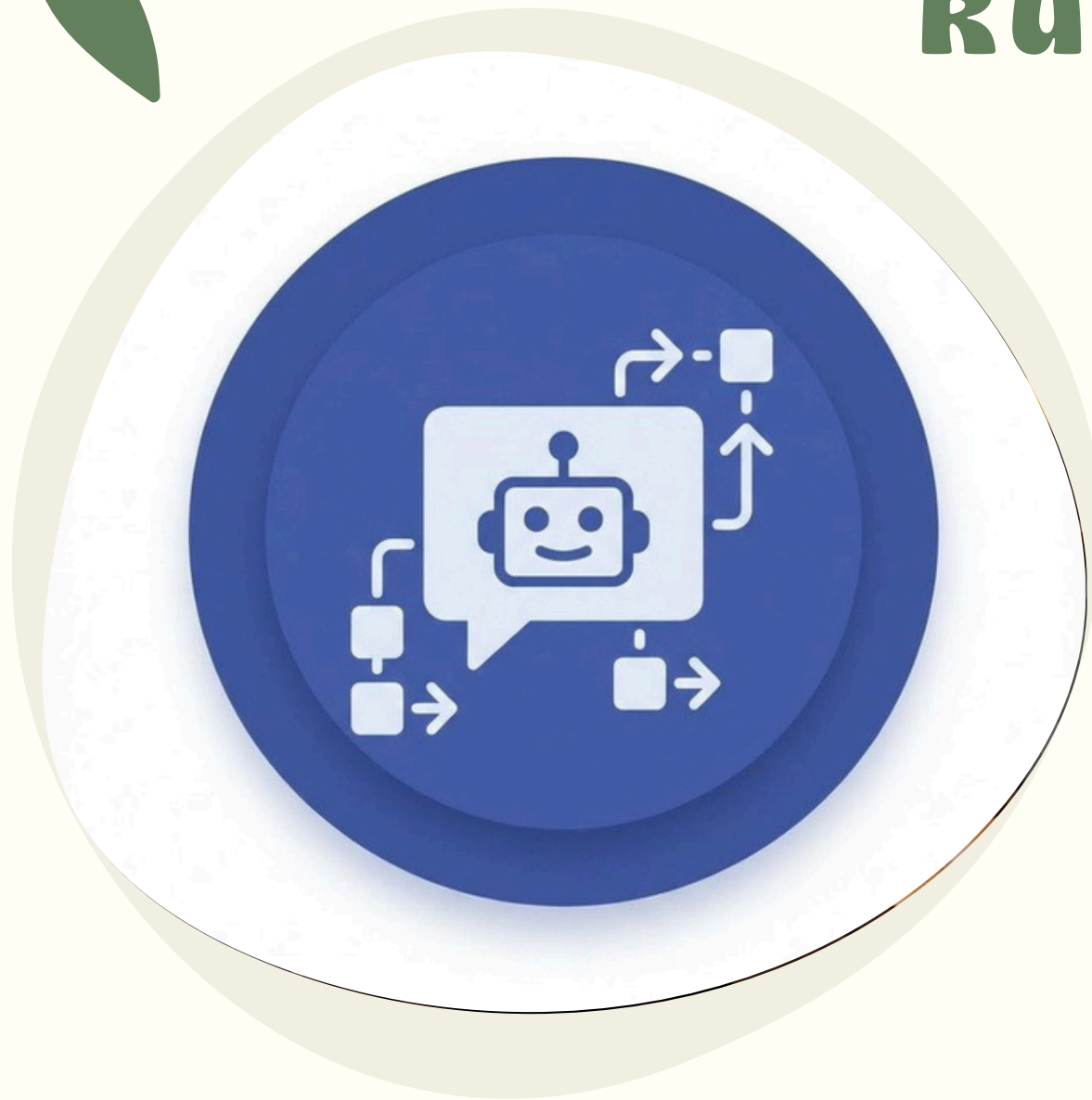
EARN YOUR BADGE 

- Requirement: Complete project 1 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 1 is the essential start for every intern.



Project 1:

Rule-Based AI Chatbot 🤖



Goal:

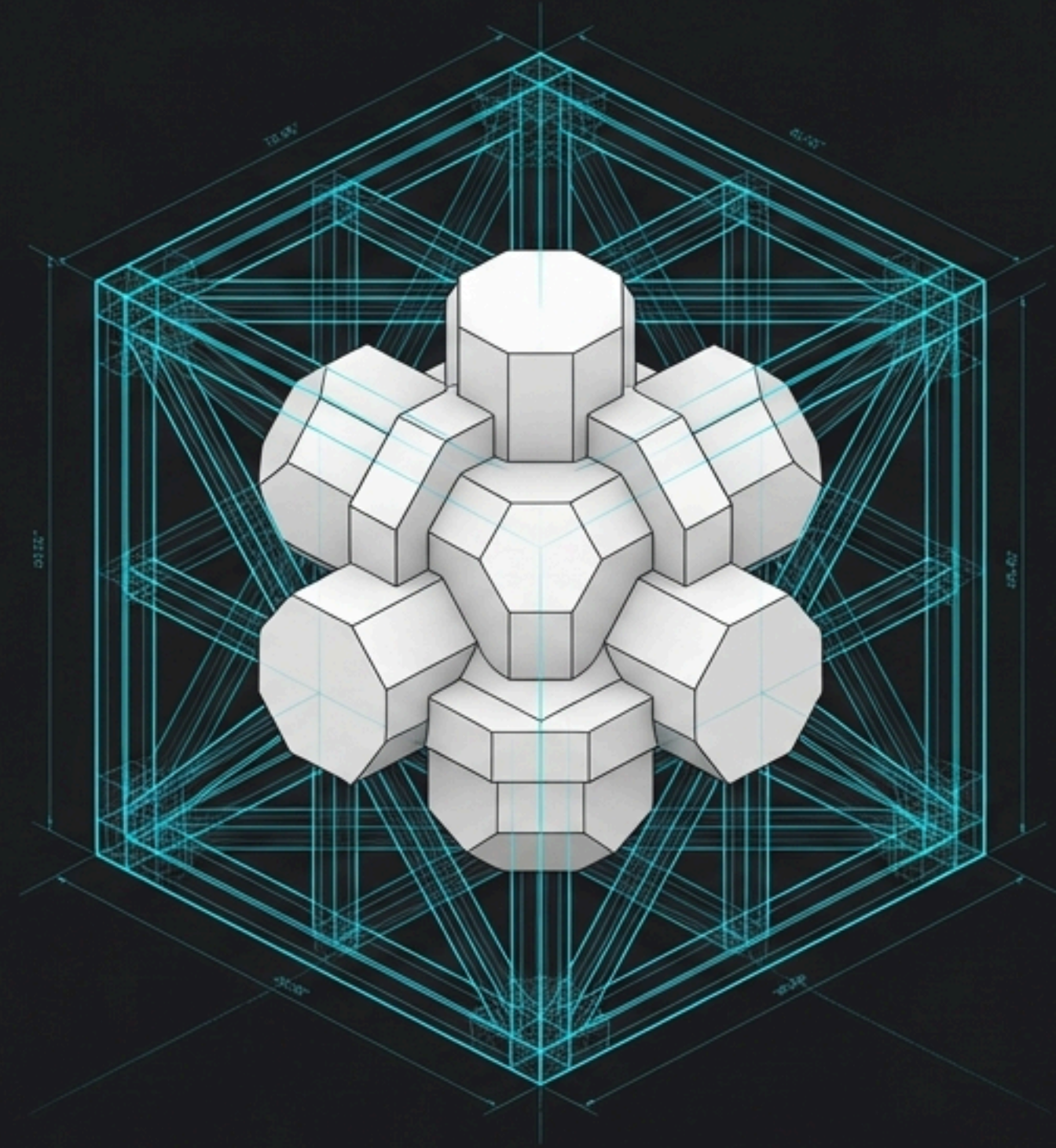
Create a simple rule-based chatbot that responds to predefined user inputs.

Key Requirements:

- Handle greetings and exit commands
 - Use if-else logic for responses
- Run in a continuous loop

Key Skills:

Control flow, decision-making logic, basic AI concepts

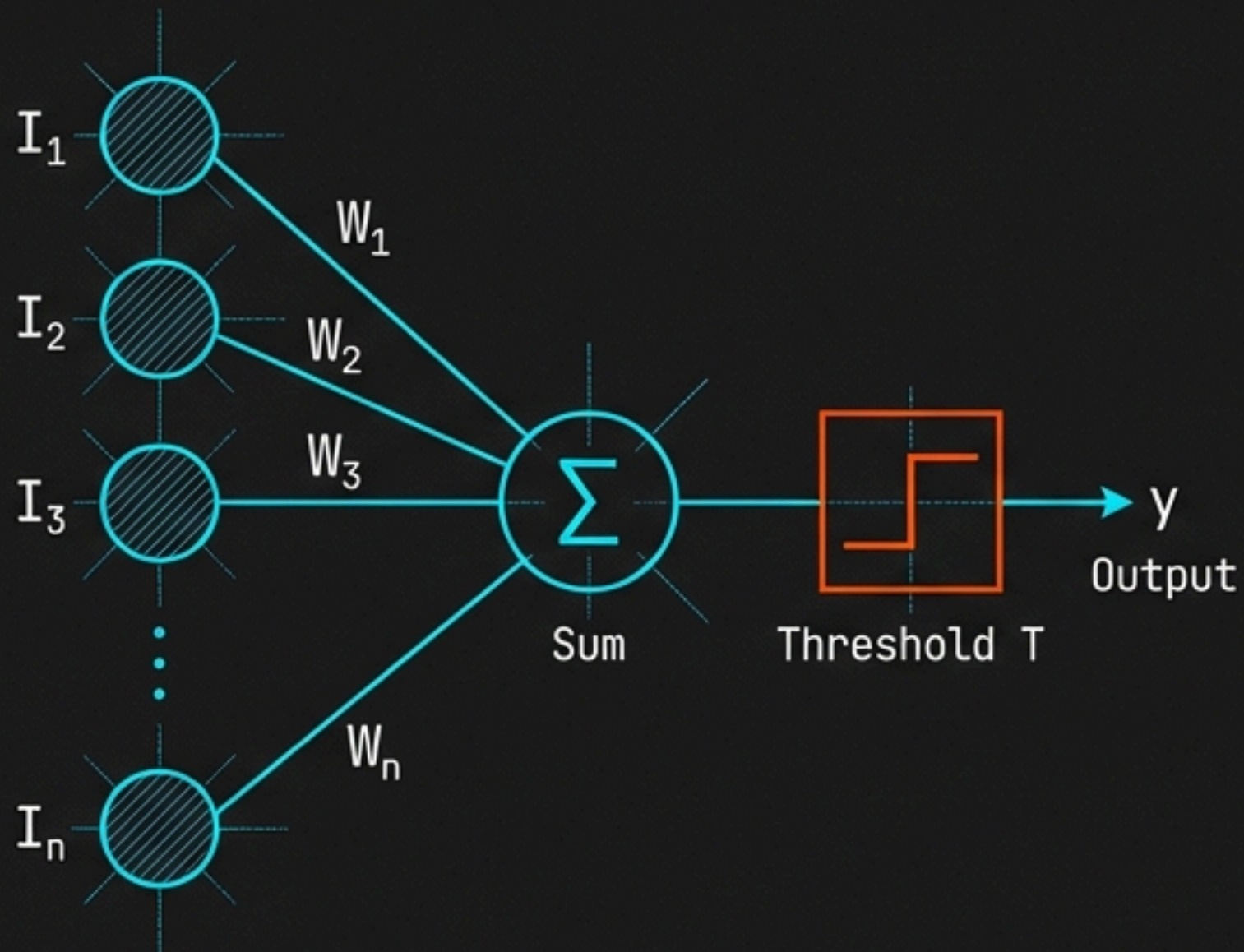


THE LOGIC ENGINE

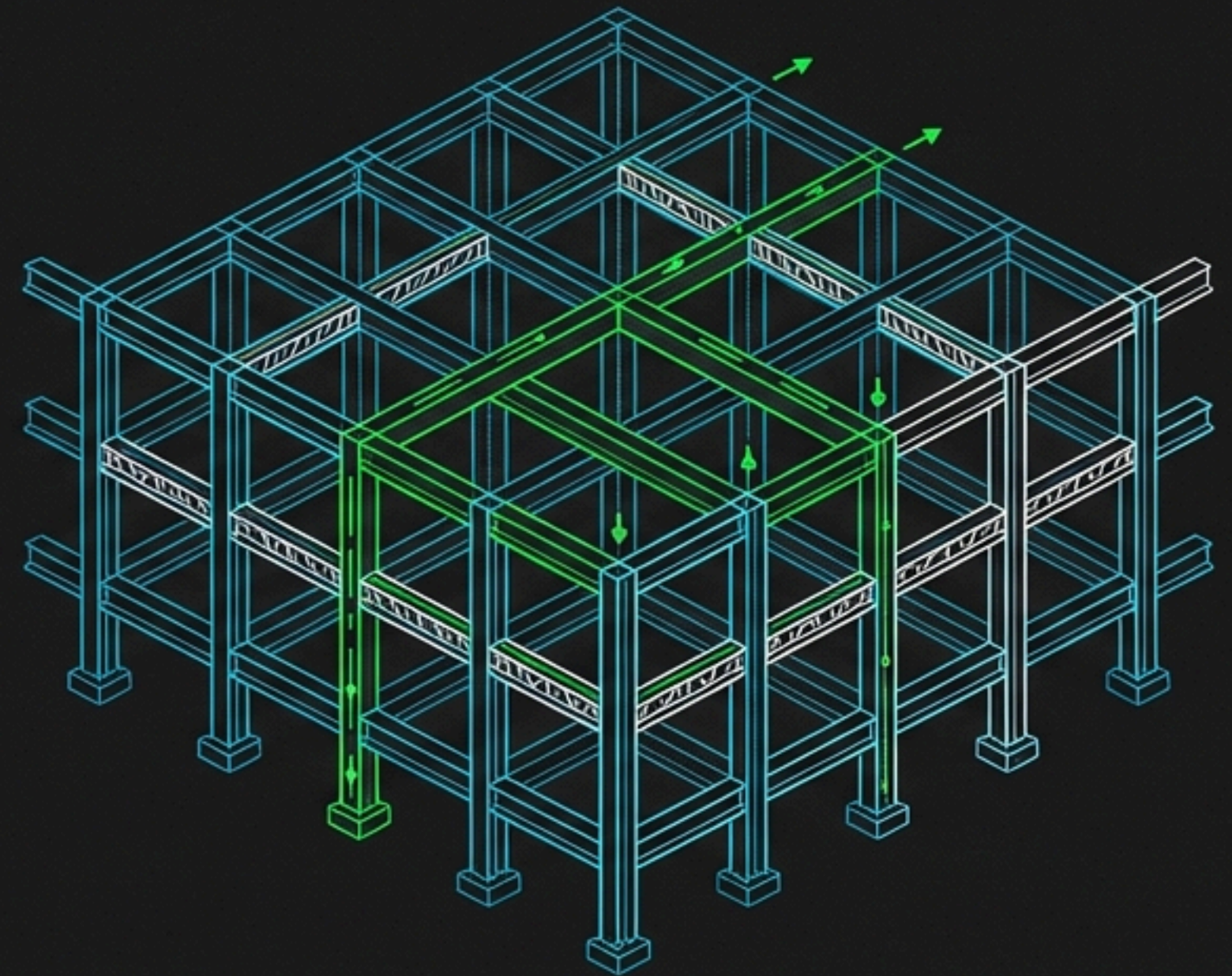
Constructing Deterministic Guardrails in the Age of Generative AI

THE TWO MINDS OF ARTIFICIAL INTELLIGENCE

SYSTEM 1: THE ARTIST (PROBABILISTIC)

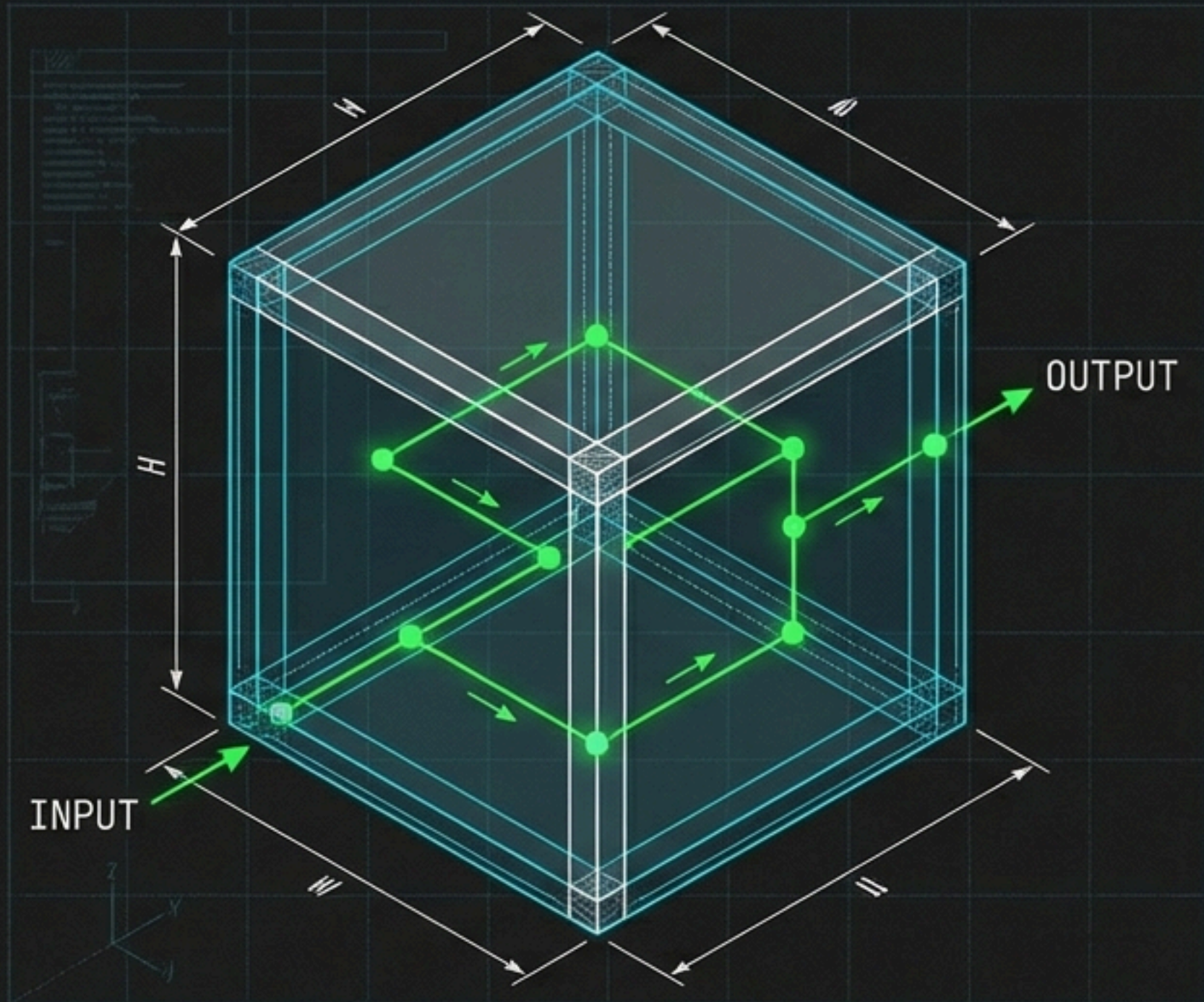


SYSTEM 2: THE ENGINEER (DETERMINISTIC)



Insight: Before you can manage the chaos of a probability engine, you must master the precision of a logic engine.

THE STRATEGIC NECESSITY OF THE 'WHITE BOX'.



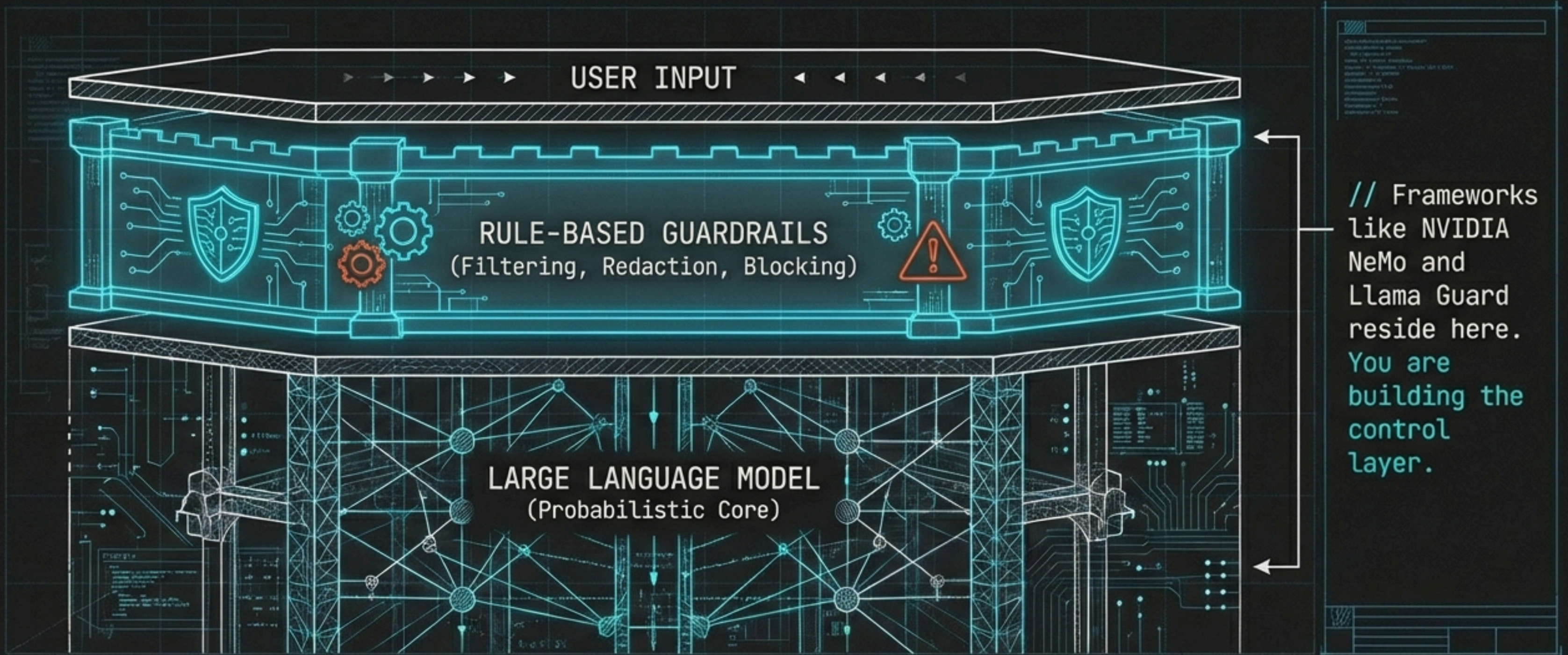
// TRACEABILITY: Input -> Logic -> Output. No mystery.

// SAFETY: Zero hallucination risk. 100% hard-coded.

// COMPLIANCE: Essential for Finance & Healthcare.

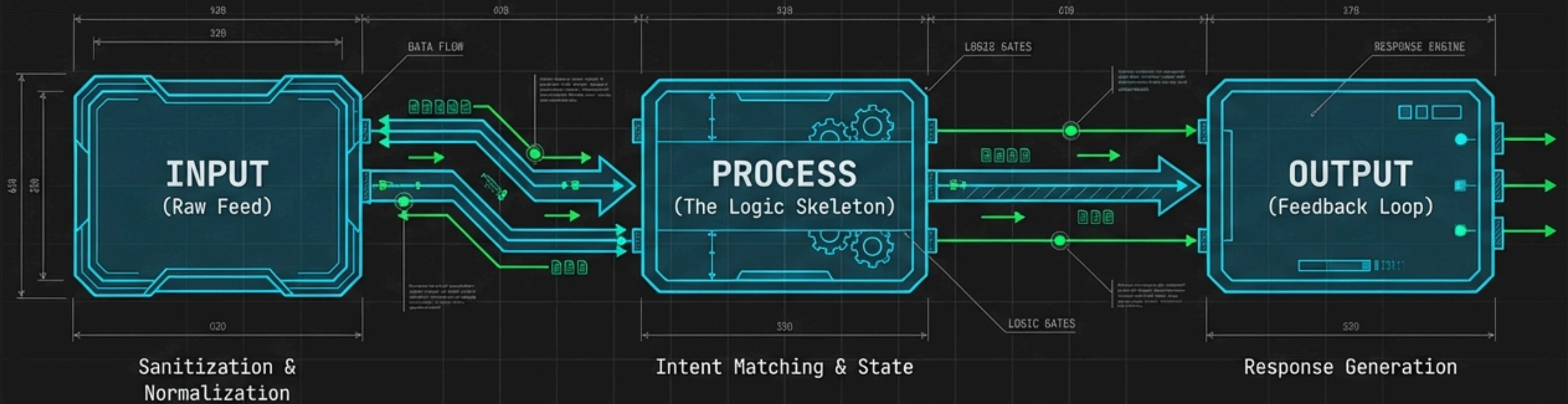
*"The rule-based generative program is a white box...
the interpreter can always provide straightforward explanations."*

THE MODERN APPLICATION: AI GUARDRAILS.

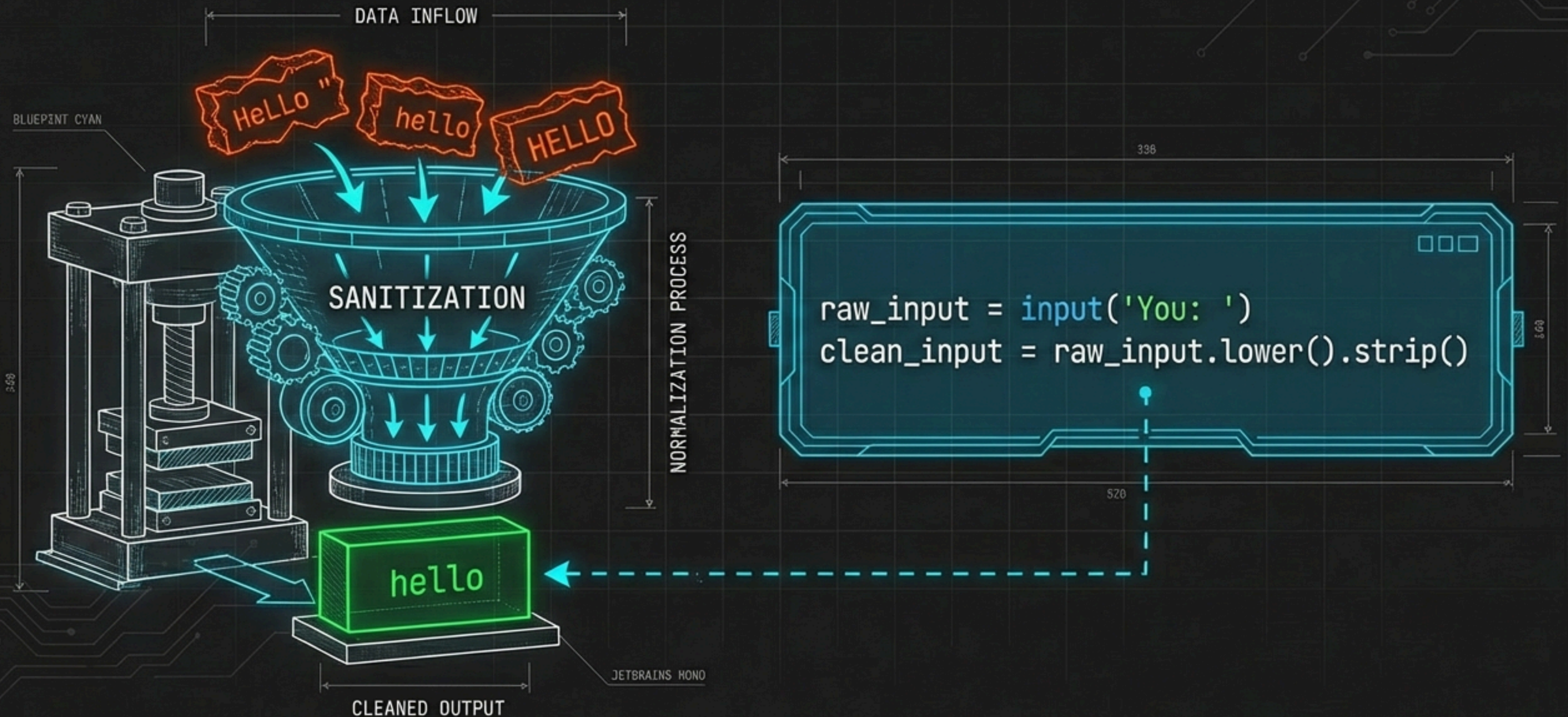


// AI guardrails act as the deterministic filter for probabilistic outputs, ensuring compliance and safety.

THE BLUEPRINT: IPO MODEL.

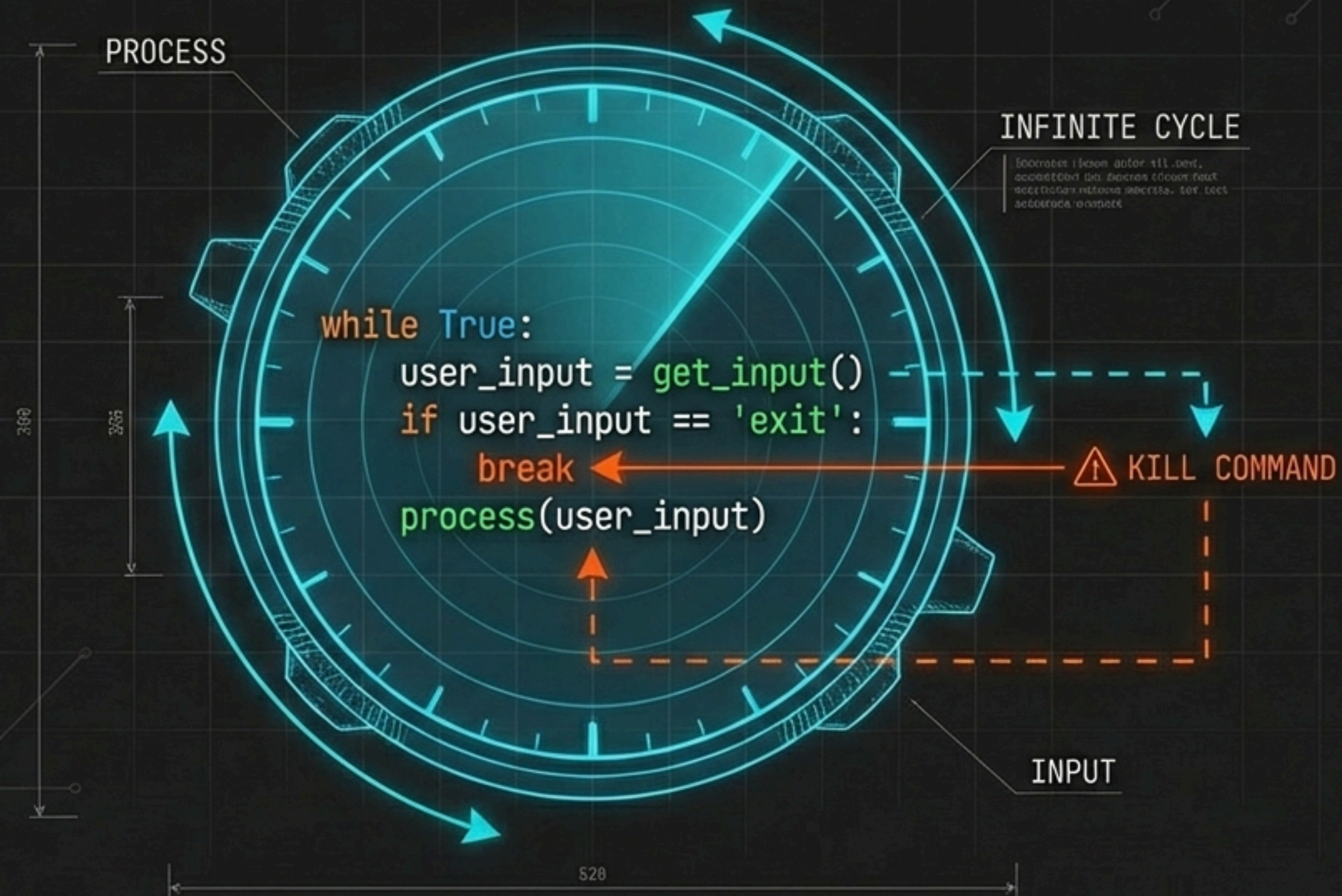


PHASE 1: INPUT & SANITIZATION.



THE HEARTBEAT: THE INFINITE LOOP.

The organism stays alive until the Kill Command.

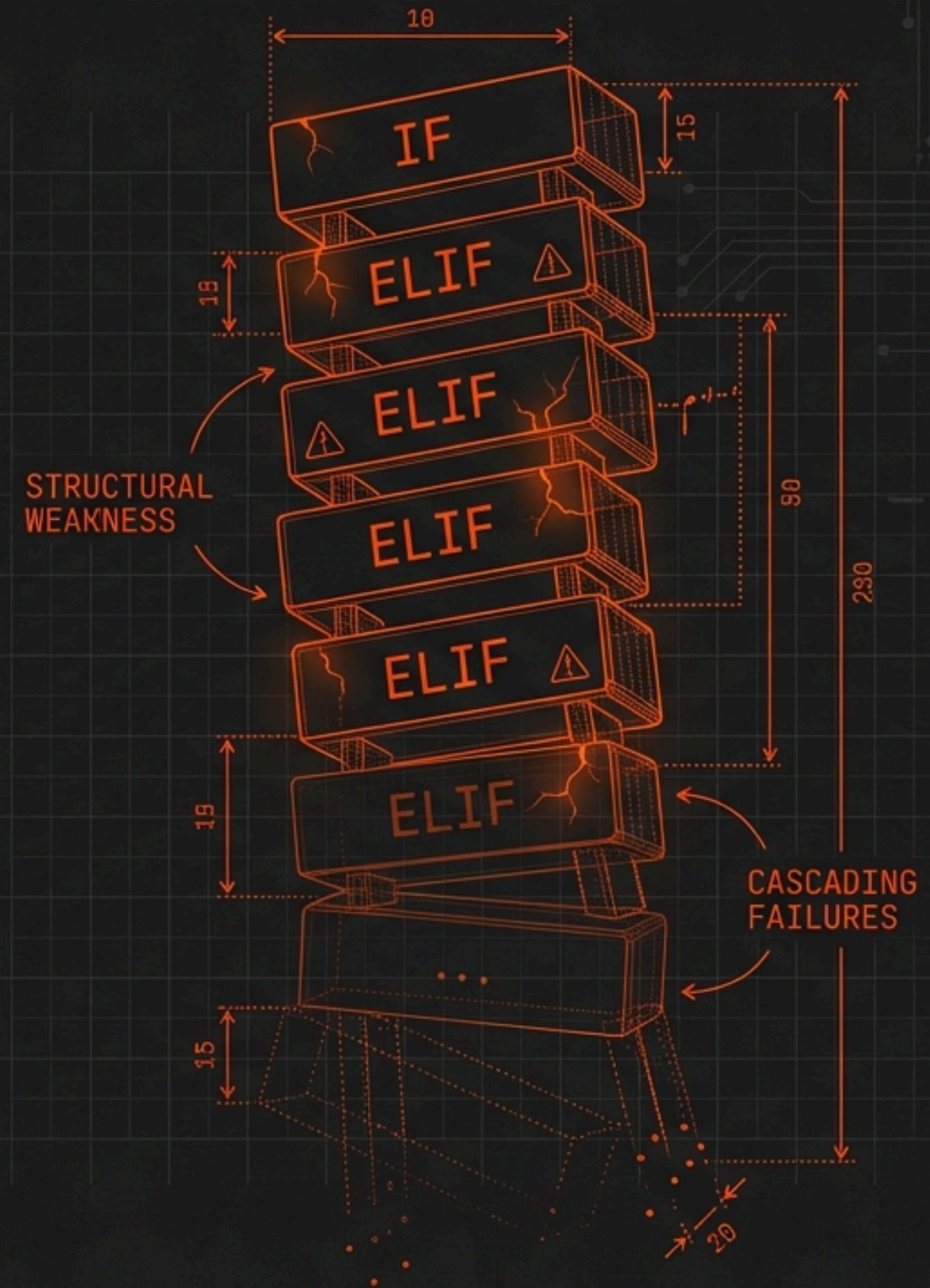


THE ANTI-PATTERN: THE IF-ELIF LADDER.

Performance: Linear Complexity $O(n)$.

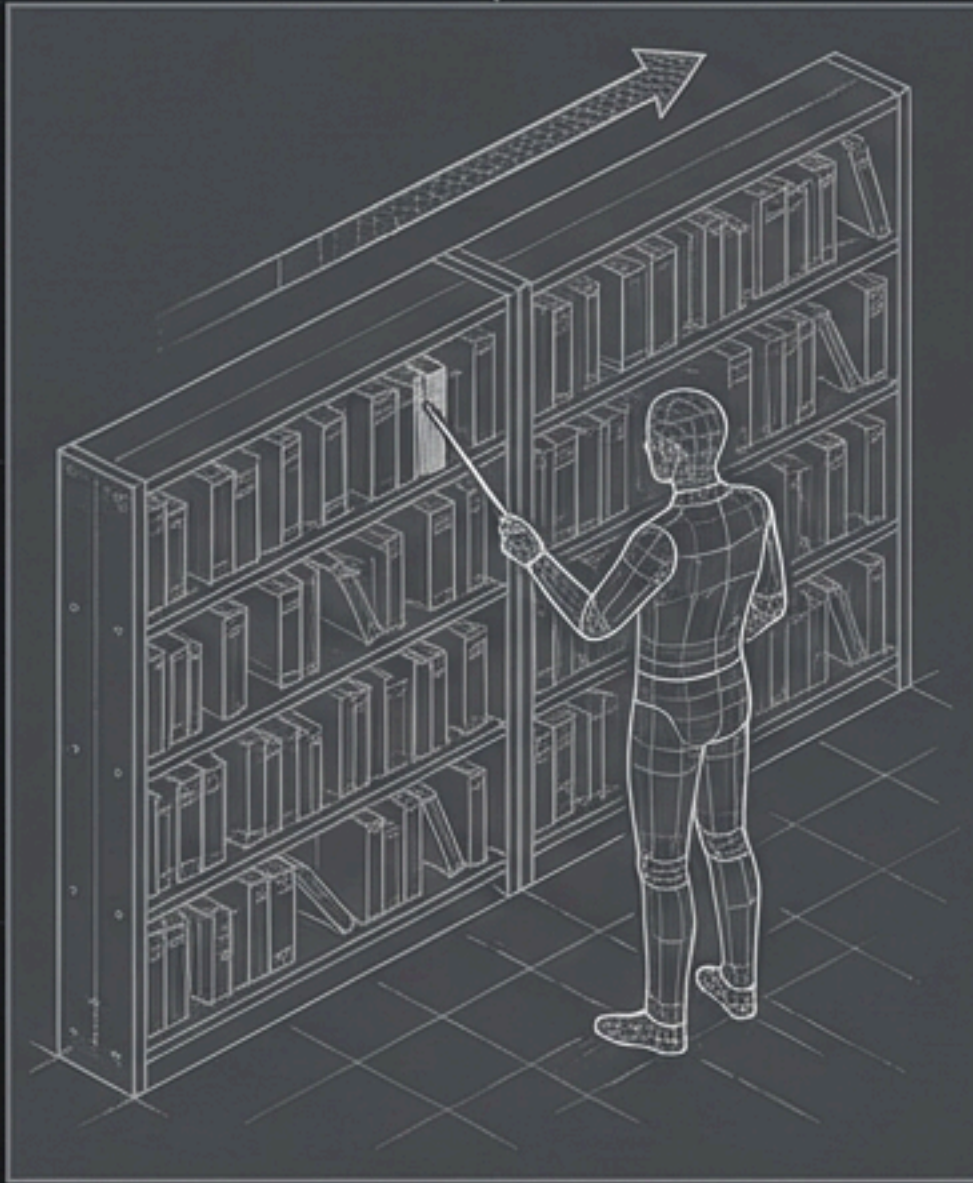
Maintenance: High Technical Debt.

Status: UNSTABLE.

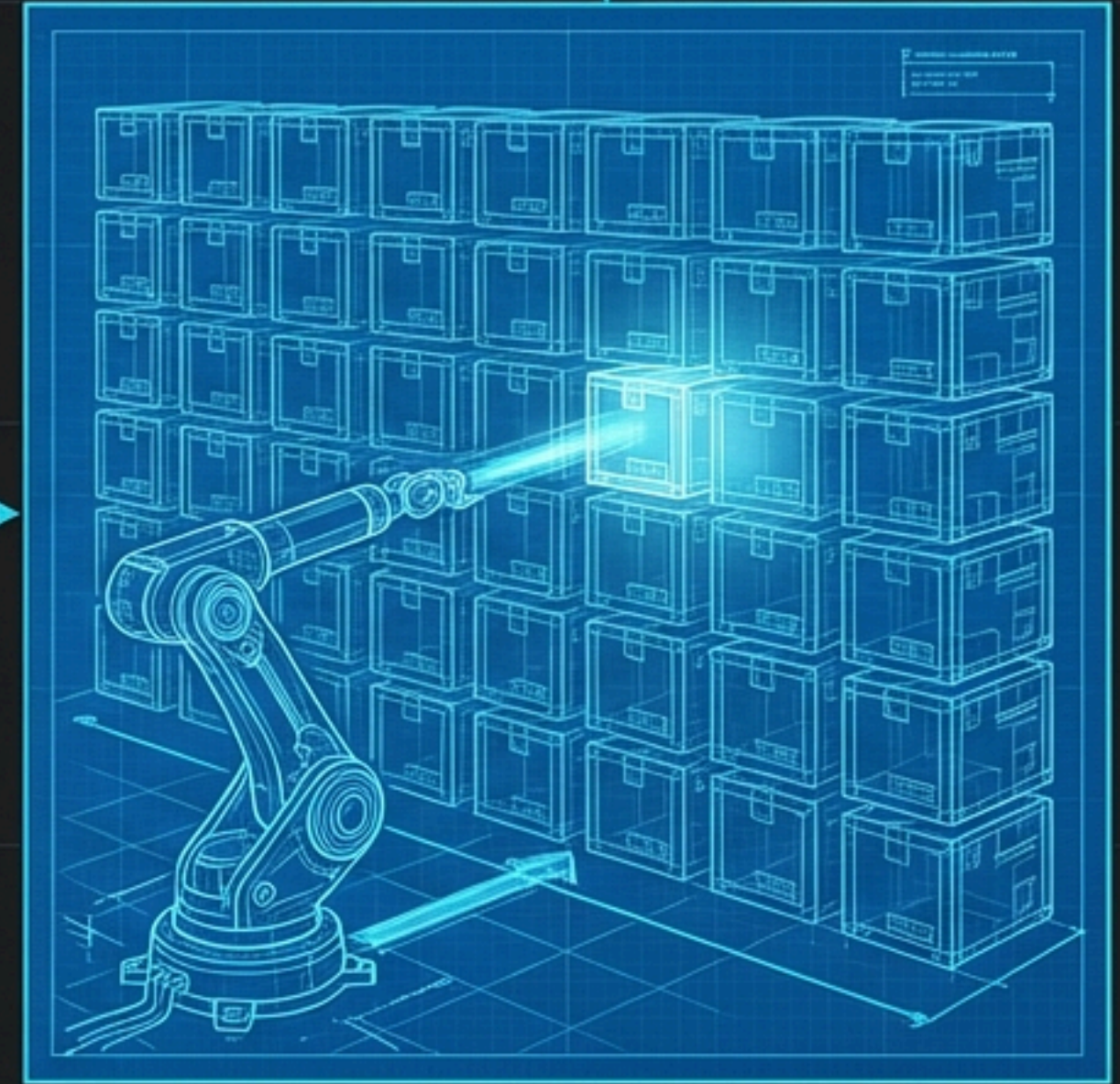
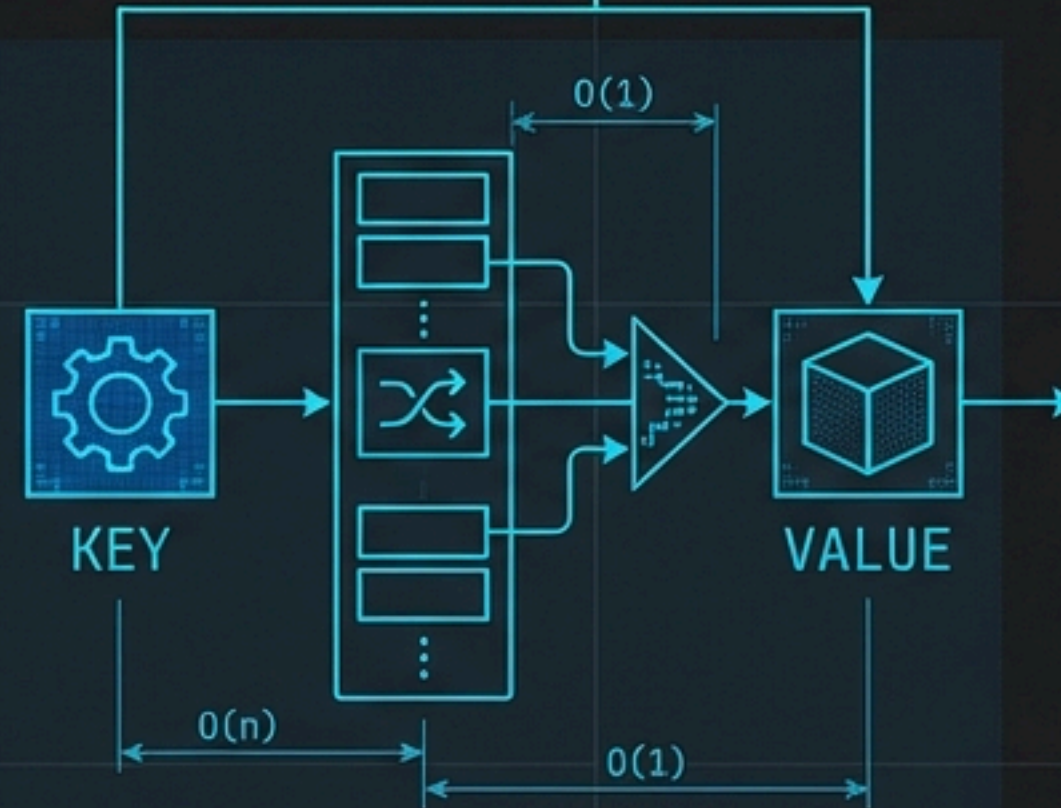


THE PIVOT: HASH MAPS & DICTIONARIES

KEY : VALUE PAIRS

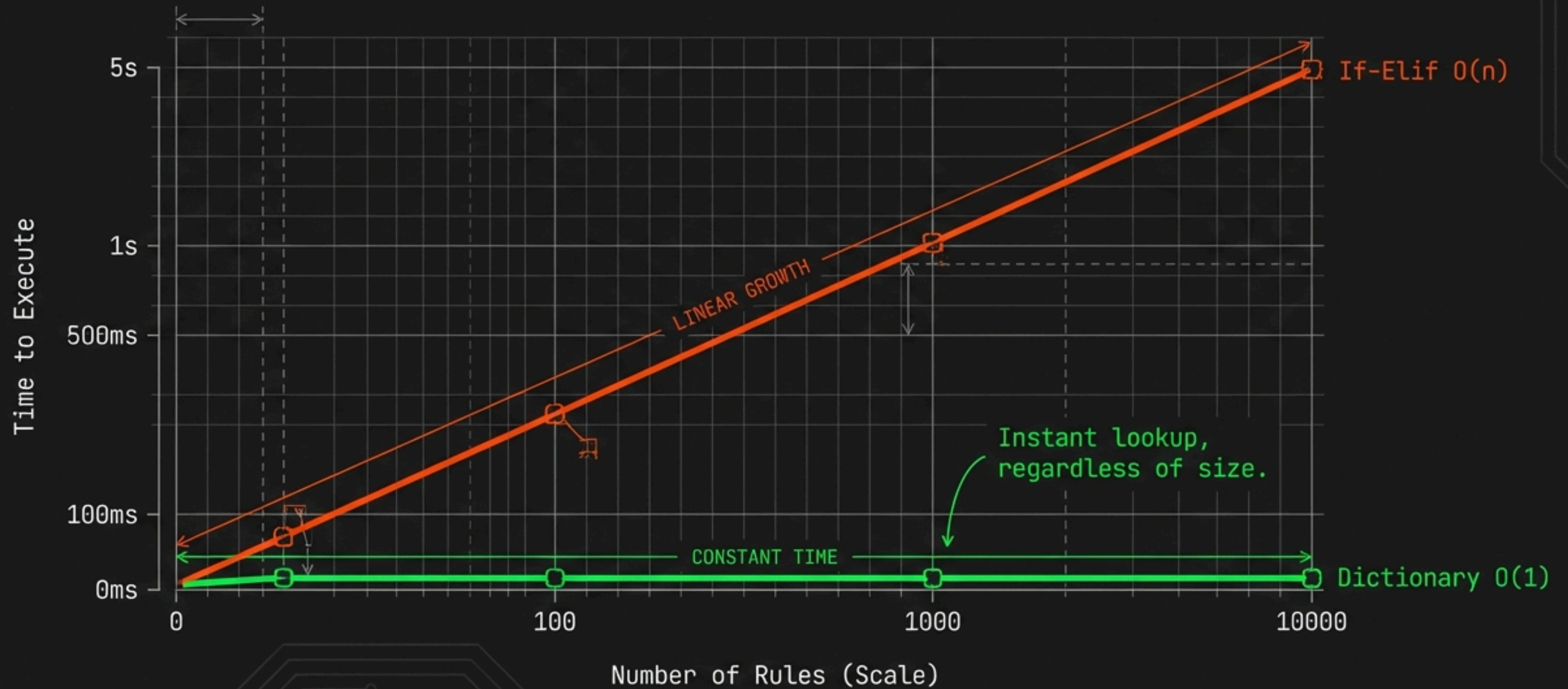


Sequential Scan



Direct Access (Hash Map)

ALGORITHMIC EFFICIENCY.



IMPLEMENTATION: THE **.GET()** METHOD.

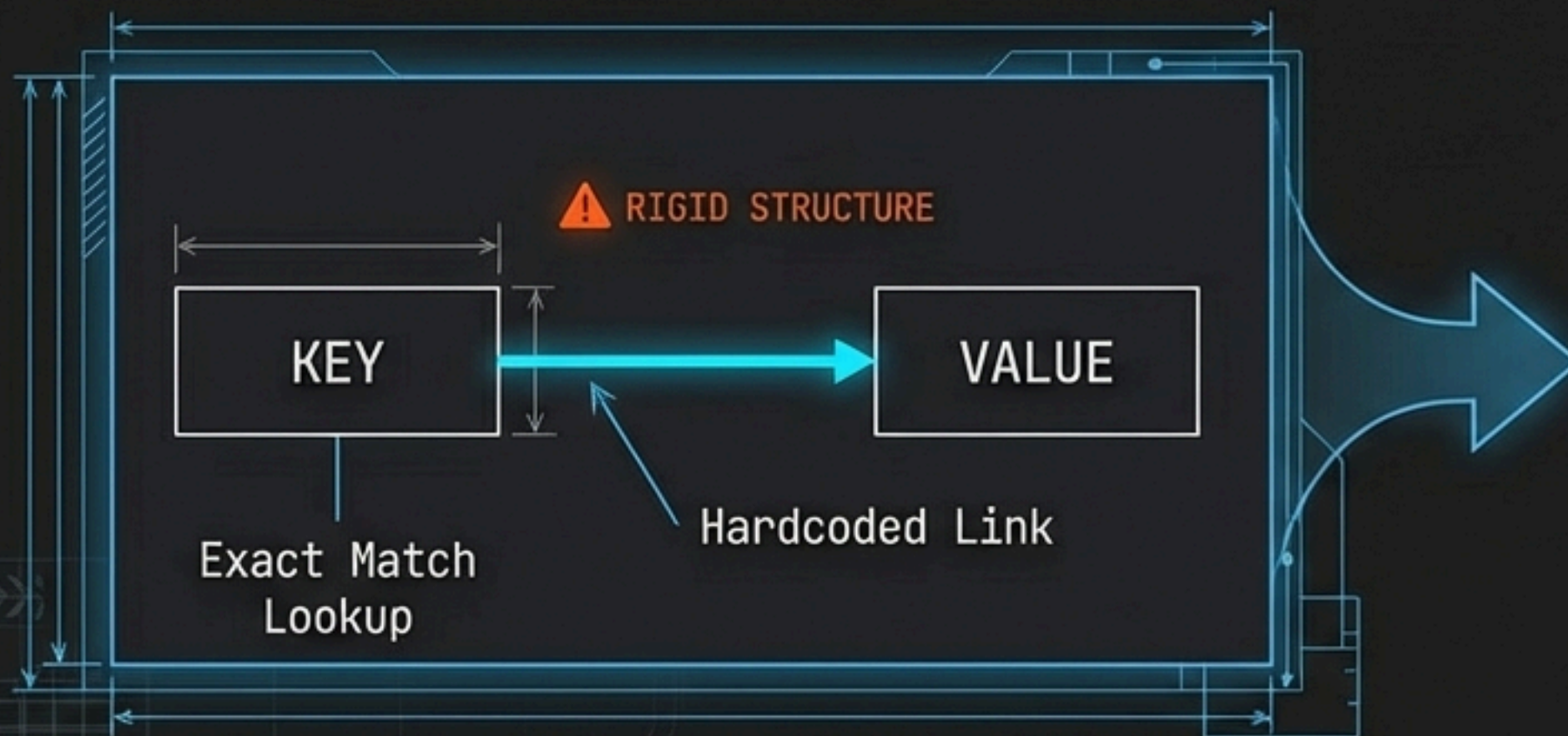
```
responses = {  
    'hello': 'Hi there!',  
    'bye': 'Goodbye!'  
}  
  
# The Professional Approach  
reply = responses.get(user_input, 'I do not understand.')
```

ATOMIC OPERATION: LOOKUP + FALLBACK

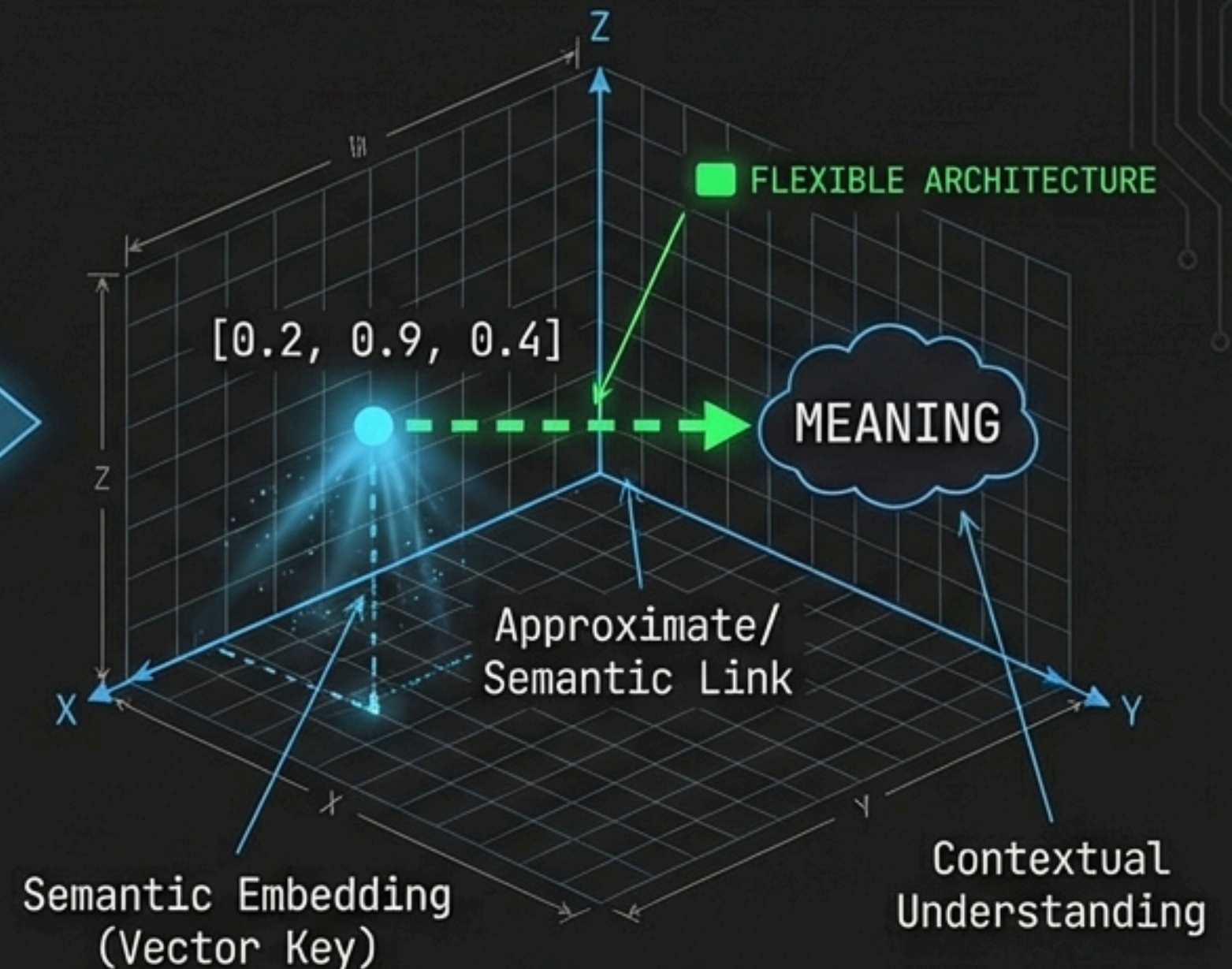
Handles lookup and fallback in a single atomic operation.

THE CONCEPTUAL BRIDGE: FROM KEYS TO VECTORS.

Project 1: Discrete Mapping (Exact Match).



Project 2: Continuous Mapping (Semantic Match).



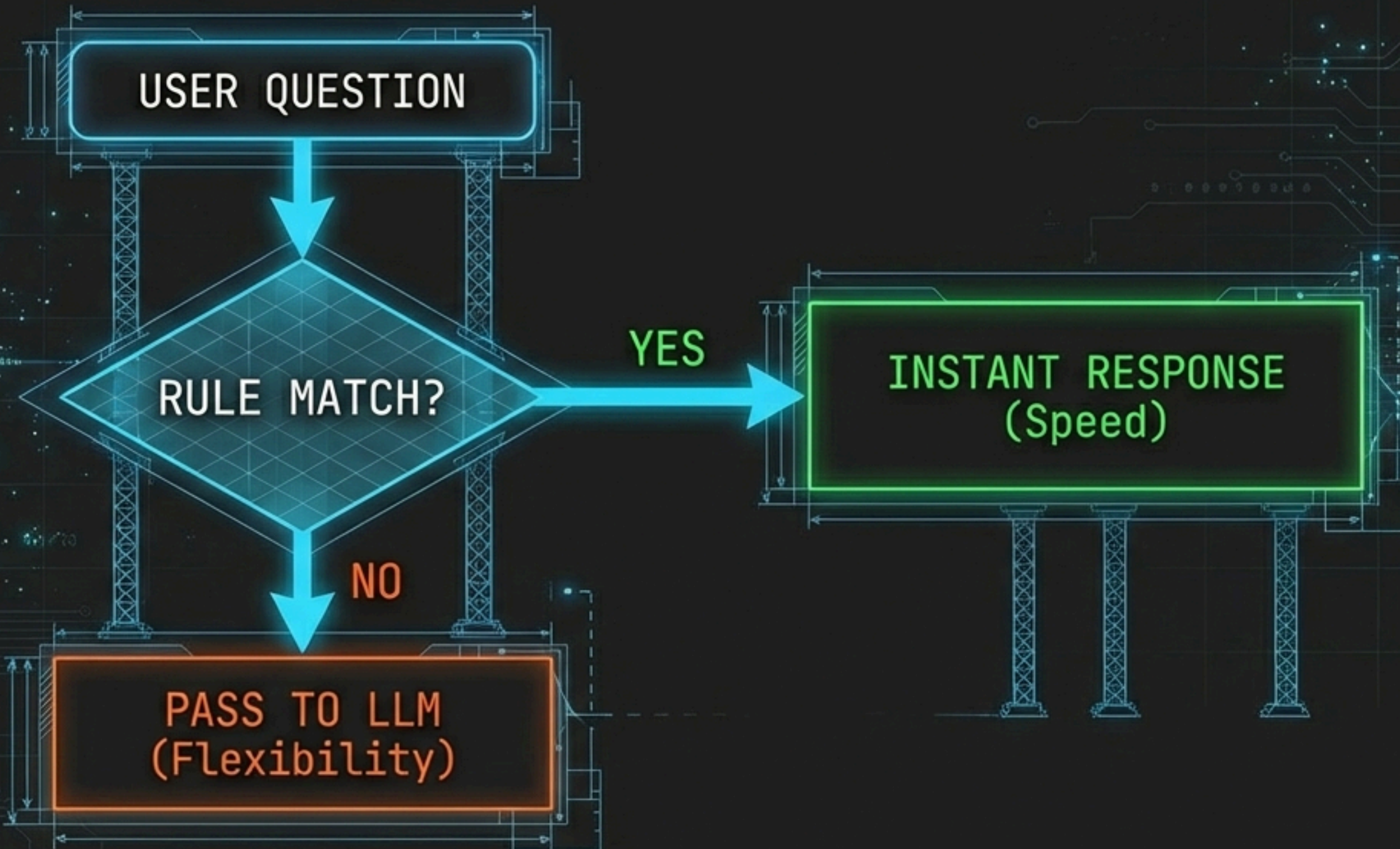
SYSTEM DIAGRAM: CONCEPTUAL TRANSFORMATION

PROJECT 1 SPECIFICATION: THE LOGIC SKELETON.

- ☐ INPUT LOOP: Continuous 'while' cycle.
- ☐ SANITIZATION: Handle case & whitespace.
- ☐ KNOWLEDGE BASE: Dictionary with 5+ intents.
- ☐ FALLBACK: Default response for unknowns.
- ☐ EXIT STRATEGY: Clean break command.

THE HYBRID ARCHITECTURE.

Rule-based reliability meets generative flexibility.





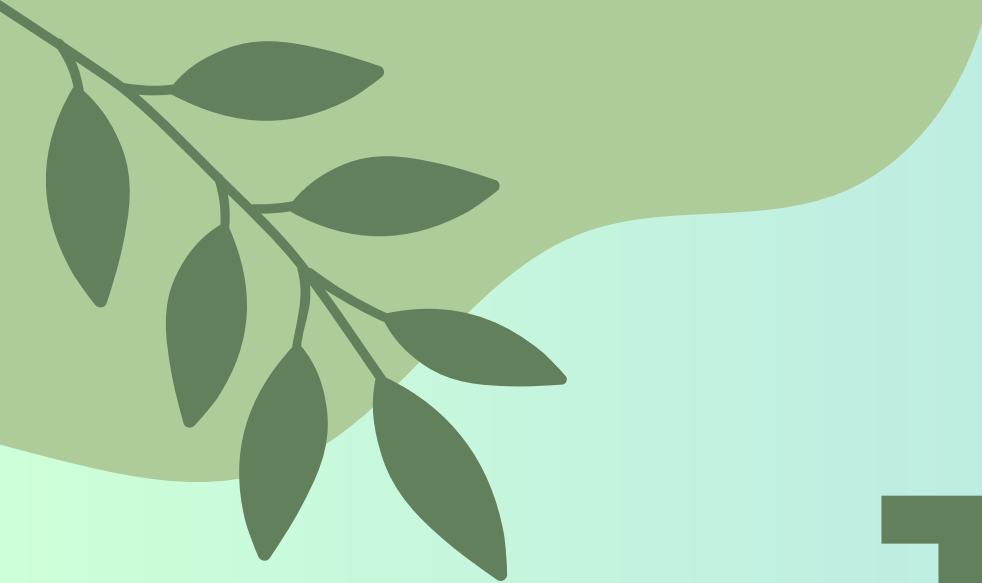
BUILD THE FOUNDATION.

An LLM without rules is a hallucination engine.
Today, we build the skeleton that holds the intelligence.

PROJECT 1 BEGINS NOW.

CONCLUSION

The absolute best way to master Artificial Intelligence is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like expanding the bot's vocabulary, adding nested conditions for smarter responses, or giving it a unique personality—and treat every unexpected output as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your grasp of foundational AI concepts to future employers. Your journey to becoming a professional AI engineer begins right here, right now, with the very first rule you define today.



THANK YOU

 +91 89330 06408

 decodelabs.tech@gmail.com

 www.decodelabs.tech

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